PHANIDHAR AKULA

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MS Computer Science • HPC, Simulation Systems & AI • Authorized to work in U.S. Oct 2026 – Oct 2029 (OPT + STEM OPT)

EDUCATION

Miami University | M.S. in Computer Science — *Aug 2024 – Aug 2026*

Oxford, OH • GPA: 3.86/4.00 • Full Scholarship • Assistantship

Thesis: SimForge: A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation (committee of three faculty advisors)

Coursework: Generative AI, Machine Learning, Advanced Database Systems, Software Quality, Software Quality Assurance, Cryptography

Malla Reddy University | B.Tech in Computer Science — *Aug 2020 – May 2024*

Hyderabad, India • CGPA: 8.2/10 • Two peer-reviewed publications (CNN soil classification, LDA topic modeling)

SELECTED PROJECTS

SimForge | Master's Thesis – A Reproducible, Cross-Simulator Benchmarking Framework for Urban Traffic Simulation — Aug *2024 – Present*

Python, OpenMP, multiprocessing, SLURM, OSC HPC (Pitzer / Cardinal / Ascend), SUMO, MATSim, DTALite, OSM, US Census PUMS

- Integrated three heterogeneous traffic engines (SUMO micro+meso, MATSim queue-mobsim, DTALite DTA) under one canonical five-file scenario bundle (network, demand, signals, config, manifest with SHA-256 hashes); **byte-identical reproducibility guarantees** across re-runs with code-enforced fair-comparison contract.
- Designed a **state-aware BFS heuristic pre-router** with OSM turn-restriction enforcement and strongly connected component (SCC) feasibility filtering, forcing engines onto byte-identical canonical routes so any cross-engine travel-time delta is provably an engine-internal property, not an input asymmetry.
- **Compressed BFS pre-routing bottleneck from ~68h to ~5–8h (~25–30× speedup)** on NYC 500K-trip scenarios via canonical-route deduplication, 16-way multiprocessing. Pool parallelism, and a JSONL content-addressed cache; deployed across three OSC HPC clusters with cluster-specific SBATCH tuning.
- Scaled to **~80K nodes / ~200K directed links** across Chicago, NYC, LA at five demand tiers (1K → 500K trips), calibrated against US Census PUMS microdata (2.4M–6.8M persons) via ModelGen, lifting realism from **~20–40% baseline to ~70–72%**. Validated by ~570 pytest tests, mutation tests on load-bearing modules, byte-identity guards, and Student's-t 95% CIs; 76% coverage.

LumiAI | Production AI Education Platform – studywithlumi.com — *Nov 2025 – Present*

React, TypeScript, Supabase (PostgreSQL/Auth/Storage), Anthropic Claude, Vercel serverless, custom domain

- pSolo-architected, built, deployed, and operate a production AI tutoring platform serving 55+ active student users; document-grounded AI chat, auto-generated quizzes and flashcards, SM-2 spaced-repetition review, and voice mode. Owned the full stack with row-level security, server-side API-key isolation, Google OAuth, and an admin analytics panel.

EXPERIENCE

Miami University | Graduate Assistant – Department of CSE — *Aug 2025 – May 2026*

Oxford, OH • Research group of Dr. DJ Rao

- Lead developer on **Cityscape** – open-source PUMS-driven population synthesizer and trip-demand generator (github.com/raodj/cityscape); contributions integrated into SimForge thesis demand pipeline. Co-administer the **Grand Challenges Scholars Program (GCSP)** mentoring undergraduate research.

Smart Pathfinders Overseas Education | Software Engineering Intern (Startup) — *May 2024 – Jul 2024*

Hyderabad, India • Internal staff/hierarchy management system (Salesforce-style CRM)

- Shipped production React/TypeScript features for an internal staff management and organizational hierarchy platform; partnered with product and design across iterative releases, addressing UI regressions and edge cases identified in post-release feedback.

Independent Clients | Freelance Software Engineer — *Aug 2023 – May 2024*

Remote • Three early-stage startup clients (under NDA)

- Delivered full-stack features end-to-end for three early-stage startup clients under real timelines; built React/TypeScript frontends and Python backends with REST APIs, MySQL, and AWS deployment.

LEADERSHIP

Graduate Students of Color Association (GSCA) | President, Miami University — *Aug 2025 – May 2026*

- Elected to lead a **104 member graduate organization** with a full executive team (VP, Treasurer, Social Chair); ran candidate speeches, elections, and ongoing operations. Organized org-wide events, coordinated cross-org collaborations, and advocated for graduate-student interests directly with the Dean and Associate Dean – **formally recognized by the Graduate School with a Certificate of Appreciation (2026)**.

Graduate Student Council Board | Elected Member, Miami University — *Aug 2025 – May 2026*

- Serve on a deliberative body reviewing student petitions and voting on academic-policy matters affecting graduate students university-wide.

TECHNICAL SKILLS

Languages: Python (advanced), C++, JavaScript, TypeScript, SQL, Java (basic)
HPC & Systems: OpenMP, multiprocessing, SLURM, OSC clusters (Pitzer / Cardinal / Ascend), Linux, Docker, Git, content-addressed caching
AI/ML: PyTorch, TensorFlow, CNNs, LDA / topic modeling, LLM API integration, document-grounded contextual prompting, agentic workflows
Web & Data: React, REST APIs, GraphQL, MySQL, PostgreSQL, AWS, OSM / Geofabrik / US Census PUMS pipelines
Engineering: pytest, mutation testing, byte-identity reproducibility guards, statistical CIs, code review, design (Adobe, Figma)

PUBLICATIONS & RECOGNITION

- Two peer-reviewed publications in IJSREM (2023): *"Topic Modelling of Web Pages with LDA Methods"* (DOI: 10.55041/IJSREM27350) · *"Soil Image Classification using Deep Learning"* (DOI: 10.55041/IJSREM23138).
- 10+ technical certifications: AWS Cloud Foundations, Applied ML & AI, Software Engineering (AICTE, Coursera, NPTEL, Forage, AWS).